

HCAHPS Patient Analysis Project

DAX Development

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Tools Used

Power BI

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DAX Development

DAX measures were developed in Power BI to calculate patient satisfaction, response rates, rankings, benchmark variance, year-over-year changes, and dynamic report information. The measures support interactive analysis across national, state, regional, measure, and reporting-period levels.

1. Average Top-box %

Create in: national_results

DAX Formula : Average Top-box % = AVERAGE('national_results'[Top-box Percentage])

Purpose: Calculates the average percentage of respondents who selected the most positive response category.

2. Average Middle-box %

Create in: national_results

DAX Formula: Average Middle-box %=AVERAGE('national_results'[Middle-box Percentage])

Purpose: Calculates the average percentage of respondents who selected the middle or neutral response category.

3. Average Bottom-box %

Create in: state_results

DAX Formula: Average Bottom-box % =AVERAGE('state_results'[Bottom-box Percentage])

Purpose: Calculates the average percentage of respondents who selected the least positive response category. It helps identify areas where patient satisfaction is low and improvement is needed.

4. National Average

Create in: national_results

DAX Formula: National Average =AVERAGE('national_results'[Top-box Percentage])

Purpose: Calculates the overall national average patient satisfaction score and serves as a benchmark for comparing state and regional performance.

5. State Average

Create in: state_results

DAX Formula: State Average =AVERAGE('state_results'[Top-box Percentage])

Purpose: Calculates the average Top-box percentage for states based on the applied filters, enabling state-level performance analysis.

6. Satisfaction Index

Create in: national_results

DAX Formula: Satisfaction Index =[Average Top-box %] + (0.5 * [Average Middle-box %])

Purpose: Calculates a weighted patient satisfaction score by combining Top-box and Middle-box percentages. It provides an overall satisfaction index for performance evaluation.

7. Response Rate

Create in: responses

DAX Formula: Response Rate =AVERAGEX('responses', IFERROR(VALUE('responses'[Response Rate (%)]), BLANK()))

Purpose: Calculates the average survey response rate by excluding invalid or non-numeric values. It helps measure patient participation in the survey.

8. State Ranking

Create in: state_results

DAX Formula: State Ranking =RANKX(ALLSELECTED('states'[State]), [State Satisfaction Score], , DESC, DENSE)

Purpose: Ranks states based on their patient satisfaction score, allowing comparison of state performance from highest to lowest.

9. Region Ranking

Create in: state_results

DAX Formula: Region Ranking =RANKX(ALLSELECTED('states'[Region]), [Regional Satisfaction Score], , DESC, DENSE)

Purpose: Ranks regions according to their average patient satisfaction score, enabling comparison of regional performance.

10. Year-over-Year Change

Create in: national_results

DAX Formula: Year-over-Year Change =
VAR CurrentPeriodDate = MAX('reports'[Start Date])
VAR PreviousPeriodDate = MAXX(FILTER(ALL('reports'), 'reports'[Start Date] < CurrentPeriodDate), 'reports'[Start Date])
VAR PreviousPeriodValue =
CALCULATE([Average Top-box %], FILTER(ALL('reports'), 'reports'[Start Date] = PreviousPeriodDate))
RETURN IF(NOT ISBLANK(PreviousPeriodDate),
[Average Top-box %] - PreviousPeriodValue, BLANK())

Purpose: Calculates the change in the Average Top-box percentage compared to the previous reporting period. Positive values indicate improvement, while negative values indicate a decline in patient satisfaction.

11. Dynamic Ranking

Create in: state_results

DAX Formula: Dynamic Ranking = SWITCH(TRUE(), ISINSCOPE('states'[State Name]),
RANKX(ALLSELECTED('states'[State Name]), [State Satisfaction Score], , DESC, DENSE),
ISINSCOPE('states'[Region]), RANKX(ALLSELECTED('states'[Region]), [Regional Satisfaction Score], , DESC, DENSE), BLANK())

Purpose: Dynamically ranks either states or regions depending on the field used in the visual. It helps create flexible ranking tables and charts.

12. Gap Analysis

Create in: state_results

DAX Formula: Gap Analysis =[State Satisfaction Score] - [Average Top-box %]

Purpose: Calculates the difference between the state satisfaction score and the national average. A positive value indicates performance above the national benchmark, while a negative value indicates performance below the benchmark.

13. Dynamic Title

Create in: state_results

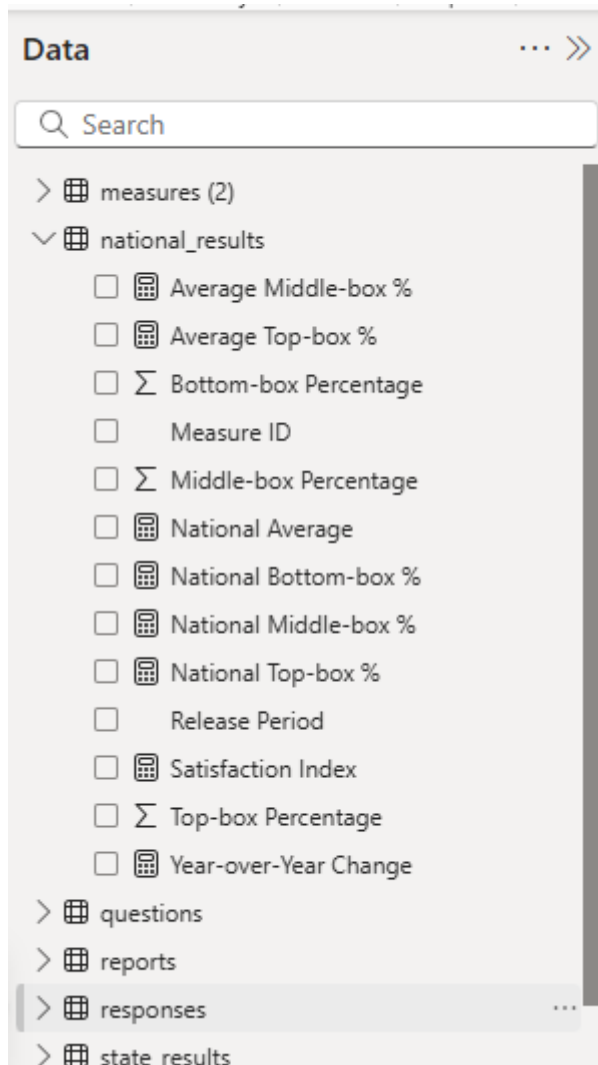
DAX Formula: Dynamic Title = VAR SelectedState = SELECTEDVALUE('states'[State Name], "All States") VAR SelectedPeriod = SELECTEDVALUE('reports'[Release Period], "All Reporting Periods") RETURN "HCAHPS Performance - " & SelectedState & " - " & SelectedPeriod

Purpose: Creates a dynamic report title that changes according to the selected state and reporting period. It helps users understand the active filter context of the dashboard.

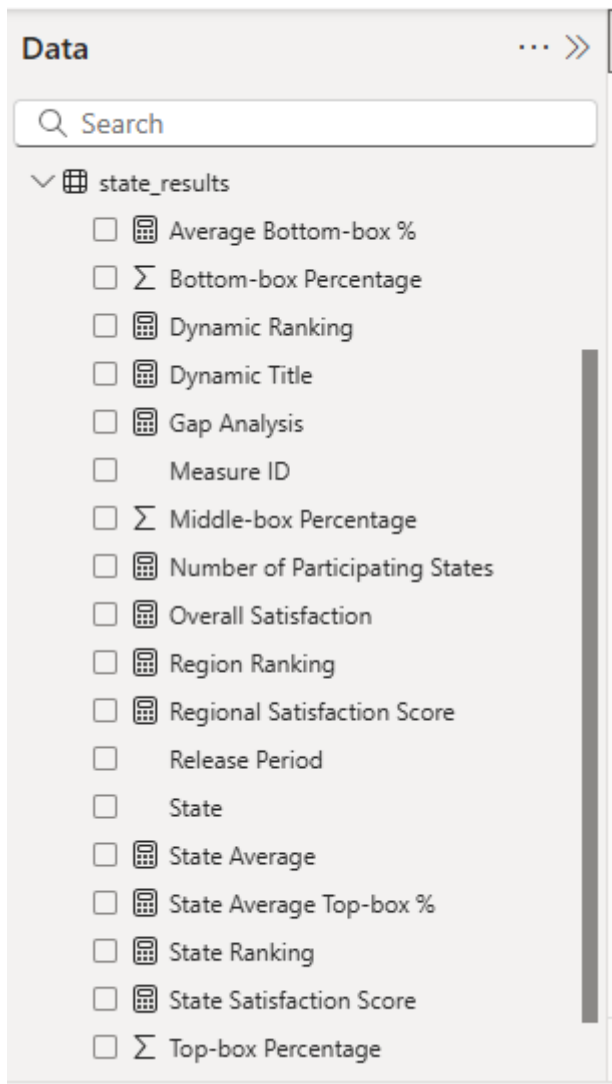
DAX Measures Screenshots

The following screenshots show the DAX measures created and implemented in Power BI Desktop.

Screenshot 1 – National Results Measures



Screenshot 2 – State Results Measures



Screenshot 3 – DAX Formula Editor

```
1 Dynamic Title =  
2 VAR SelectedState =  
3     SELECTEDVALUE('states'[State Name], "All States")  
4  
5 VAR SelectedPeriod =  
6     SELECTEDVALUE('reports'[Release Period], "All Reporting Periods")  
7  
8 RETURN  
9 "HCAHPS Performance - "  
10 & SelectedState  
11 & " - "  
12 & SelectedPeriod
```

Conclusion

The DAX measures developed in Power BI provide dynamic calculations for patient satisfaction, benchmarking, rankings, response rates, trend analysis, and interactive reporting. These measures improve analytical capabilities and support informed decision-making through responsive dashboard visualizations.